Nicholas Enrique Selvaraj

Singapore | +65 9037 0150 | nicholasenrique09@gmail.com | linkedin.com/in/nicholas-enrique-selvaraj

SUMMARY

Year 3 Engineering Product Development student at Singapore University of Technology & Design (SUTD), with a Diploma in Aeronautical Engineering and two years as an RSAF medic. Award-winning project work in healthcare device design and accessibility, using Fusion 360 CAD, FEA, and 3D-printed prototyping. Seeking internship in [medical device development / aerospace engineering].

SKILLS

CAD & Simulation: Fusion 360, AutoCAD, FEA (structural)
Programming & Analysis: MATLAB, Simulink, [Python]
Prototyping: FDM 3D printing, [laser cutting], iterative prototyping, sustainable material selection
Methods: Design thinking, user-centred design, user research, Agile workflow, cross-functional coordination

EDUCATION

Singapore University of Technology and Design (SUTD) Sep 2024 – Present
B.Eng. Engineering Product Development; Minor in Design, Technology & Society | GPA 4.43/5.00

- Best Biology × Design Thinking Project (Term 3); Best Design Thinking & Innovation Project and Sustainability Practice Prize (Term 2); Top 10 Finalist, Design Thinking Project (Term 1).

Singapore Polytechnic 2018 – 2021
Diploma in Aeronautical Engineering with Diploma Plus in Aviation Management | GPA 3.667/4.00

- Lee Foundation Bursary Award, 2019.

DESIGN PROJECTS

RackIt — RFID Smart Toolrack, SUTD [Term 2, 2025]

- Integrated RFID tracking with an on-screen assistant to automate tool identification, real-time inventory and step-by-step usage guidance for students.
- Built the enclosure entirely from sustainable materials; won Best Design Thinking & Innovation Project and the Sustainability Practice Prize.

Can Remember or Not? — Alzheimer's Memory Engagement Kit, SUTD [Term 3, 2025]

- Designed a 3D-printed tactile pop-up scrapbook using Singapore kopitiam imagery, paired with a 100% recycled-material game kit, to support memory recall in Alzheimer's patients.
- Won Best Biology × Design Thinking Project; selected for the DTP III Exhibition.

MakanMap — Accessible Hawker Kiosk for the Visually Impaired, SUTD [Term 4, 2026]

- Designed a kiosk with 3D-printed food-shaped tactile buttons and audio menu feedback, plus a dynamic mechanical-pin tactile map directing users to their chosen stall.
- Iterated button geometry across 10 printed versions, validated through testing with visually impaired users.

CareWeave — NICU Milk Bottle Trolley, Universitas Airlangga [Jan 2026]

- Co-developed a neonatal milk-handling trolley with a cross-institution team in Surabaya under SUTD's Global Exchange Opportunity (GEO) programme; placed 2nd for Best Product and Innovation.

Guard-Damp-It! — Intelligent Safety-Assisted Weedwhacker, SUTD **[**Term 5, 2026]

- Co-developed a smart weedwhacker guard that dampens projectiles and encourages safer operator technique; achieved 61% projectile energy dissipation and commended by industry partner and faculty reviewers.

WORK EXPERIENCE

Part-Time Associate | Ministry of Football Aug 2025 – Present

- Coordinate multi-school sports sessions, competitions and contingency activities across [X] schools on short lead times.
- Liaise with MOE educators to plan training programmes and manage equipment logistics across concurrent sites.

Medic, Full-Time National Service | Republic of Singapore Air Force Oct 2022 – Sep 2024

- Managed medical centre operations including medication distribution, inventory control and regulatory compliance.
- Delivered frontline medical coverage at RSAF Open House 2023 for 5,000+ visitors in an active aerospace environment.

Healthcare Support Assistant, Part-Time | Fullerton Health Group Jul 2021 – Oct 2021

- Coordinated vaccination workflow across patient ushering, registration and post-vaccination monitoring under evolving MOH COVID-19 protocols.